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# Dance Guiding Art Innovation Framework Based on Video Action Recognition Technology

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***Abstract*—For dance, it can not only improve people's aesthetic ability and self-quality, but also benefit people's physical and mental health and physical fitness. Therefore, in the process of college dance teaching, teachers should increase the importance of dance teaching, formulate a set of teaching plans that are in line with the development of students according to the actual situation of students, and continuously cultivate students' dance awareness. The main research is to use image preprocessing, 3DZernike moments, building codebooks and support vector machines (SVM) to recognize human actions in action videos. Simulation test results show that this method can effectively identify KTH and A-go-go dance video databases.**

***Keywords—Video Action Recognition, Dance Education, Art Innovation, Innovation framework***

## I. INTRODUCTION

In recent years, action recognition has become a hot research topic in the field of computer vision, and has great research value. Scholars and research institutions at home and abroad have done many studies and achieved good results [1-3]. However, there are relatively few research results on applying motion recognition technology to dance videos, mainly because the complexity of human movements in dance videos is relatively high. At the same time, there is a problem of self-occlusion in dance, which leads to low accuracy of action recognition [1-6].

Therefore, the development of motion detection and recognition in dance videos is relatively slow. Based on the problems mentioned, this paper proposes a dance video action recognition method based on computer vision. This method firstly performs gray-scale processing on the collected color images, thereby reducing the amount of computer calculations. Then use the Gaussian mixture model to eliminate the background of the generated grayscale image to obtain a black and white noise-containing video image. After median filtering and denoising, the foreground target in the image is obtained. The related research on the combination of video action recognition technology and dance art has just started in China. By applying human motion recognition technology to dance videos, dance motion gestures can be effectively recognized. By comparing the video action with the standard action, the dancer's dance posture can be evaluated, and suggestions for modification can be given. It is an advanced auxiliary training method [7-12].

This paper adopts the related technology of video human action recognition, first trains an effective SVM classifier with a relatively simple KTH database, and then uses a more complex A-go-go dance video database to intensively train the SVM classifier, and finally uses training the obtained mature classifier recognizes A-go-go dance videos and obtains a good classification effect. The action recognition in this paper is one of the most popular research directions in the field of computer vision research, and its application range has covered intelligence Human-computer interaction, virtual reality, motion-assisted analysis, video surveillance, video retrieval, etc., especially in the field of virtual reality, there have been a large number of application results based on motion recognition, which has brought subversive changes to some scene applications in our lives . However, there are still few researches on motion recognition based on the combination of video-based motion recognition technology and dance video, and due to the high complexity of dance moves and the self-occlusion of dance moves, the research on dance video-based motion recognition needs to be further develop [13-17].

The successful application of action recognition technology in other fields also provides us with sufficient theoretical basis for applying it to dance video action recognition. In view of the large amount of music and dance video materials that currently exist, professionals need to spend a lot of time listening to and seemingly analyzing these dance video materials. This method is undoubtedly extremely inefficient. If the motion recognition technology is applied to the analysis of these music and dance videos to obtain organically connected music-dance action clips, it will not only reduce the work intensity of dance professionals, facilitate the retrieval of music and dance video data, but also enable an automatic dance choreography system More efficient, and the results obtained are more colorful [18-20].

In the course of college dance teaching, teachers should continuously improve and innovate teaching methods if they want to continuously cultivate students' dance awareness. For example, teachers can integrate music into dance teaching, use music perception and listening, to continuously cultivate students' dance awareness and promote students' future development and progress. In the process of dance teaching, through listening to music, students can have a preliminary understanding of the melody and rhythm of music. In this way, students can deepen their understanding and cognition of dance movements, and continuously cultivate their dance awareness, thereby comprehensively improving their own dance level. In addition, students should deepen their understanding and analysis of music in the actual dance learning process. Music can bring us auditory feelings and inner resonance. Therefore, in the actual dance teaching, we should appropriately incorporate music, and look for the essence of music from the structure, theme, climax and style


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of the music, so as to accurately choose the dance action. Ways of expression to improve students' dance level [21-22].

The IST organization from the European Union led the ADVISER project in 1999, which mainly conducts research on human motion analysis and human-computer interaction. The intelligent video surveillance system developed by the University of Maryland overcomes the interference of scene lighting changes, can automatically detect human targets in the video [23-24].

## II. The Proposed Methodology

### A. Video Action Recognition

Before digital calculation and recognition of video images, it is necessary to perform preprocessing such as gray scale transformation, image thresholding and image segmentation to reduce the amount of computer calculations and extract effective information. Usually, the video image is originally a true color image, that is, an RGB image, and each pixel of it is composed of the three primary colors of R, G, and B. Because of the huge amount of calculation to directly process the true color image, the analysis efficiency is low. Therefore, this article uses the gray-scale transformation method to process the video image into a gray-scale image to reduce the color information contained in the image. The gray-scale image is shown in Figure 1.

Fig. 1. The original image and the image after gray scale transformation

The purpose of image thresholding is to obtain a binary image of a moving image. The main realization method is to select an appropriate threshold, use the threshold to distinguish the standard, and then divide the pixel gray value into two categories, so as to realize the segmentation of the moving image. The threshold can generally be written as follows:

$$T=T[x, y, f(x, y), p(x, y)] \tag{1}$$

After the binary image of the current moment of the action video is obtained, the motion area needs to be separated from the scene, which involves segmentation of the motion area. In this paper, the binary image processing function of Matlab software is used to find the edge of the human body contour in motion by setting an appropriate threshold. The specific method is: suppose the current time is t, the frame of the detection video at time t is P(x,y,t), the size is M×N, and the binary image obtained is A(x,y,t). At time t, the foreground gray value of the binarized image A(x,y,t) is 255, and the background gray value is 0. Set to scan along the column vector of A, and count the number C of foreground pixels in each column (1<C<n). The maximum value Ci is selected from the n frequency data, and the column serial number corresponding to the maximum value is calibrated as i. If the ratio of the number of lines between Ci and the frame m is greater than 16.

### B. Dance Education

Dance is a relatively important part of university education. It can not only help students strengthen their bodies and cultivate their sentiments, but also cultivate their social communication skills and self-quality to a large extent. For dance teaching in major colleges and universities, the purpose is to make the form of dance have a certain collective character. In the process of dance teaching, teachers can reasonably combine the inside and outside of class, truly fully express social life and study life, and constantly cultivate students' dance consciousness. Dance is a very comprehensive art. When learning dance, students not only need to have very good appearance conditions, but also have a high cultural level and artistic accomplishment. Teachers should properly carry out some dance activities when teaching, so that not only can develop students' intelligence, but also some potential non-intelligence factors of students. Therefore, in the process of college dance teaching, teachers must increase the importance of students' dance awareness, so as to lay a solid foundation for the future development of students.

Fig. 2. Dancing tutorial

Since the beginning of the 21st century, society's demand for talents has become higher and higher. Therefore, in the process of college dance teaching, teachers should continue to reform and innovate the dance teaching system, actively introduce advanced teaching models, and improve and innovate the traditional dance teaching models to greatly improve the innovative training of students' dance awareness.

$$\cos(x, y) = x^T y / [(x^T x)(y^T y)]^{1/2} \tag{2}$$

In this way, students can better appreciate the meaning of dance, and more deeply appreciate the charm and value of dance art. In recent years, more and more dance teachers have


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devoted themselves to the reform of the teaching system, increasing the emphasis on the cultivation of dance consciousness and innovation ability, so as to promote the future development of students. However, due to the continuous development of the times, there will still be some minor problems in the actual dance teaching process, which affect the cultivation of students' creative ability in dance consciousness.

### C. Dance Education Art Innovation

The new dance art vocational education implements a project-oriented teaching method, which is a successful attempt of Shanxi Vocational Art College. The so-called project-oriented classroom teaching is to take the project as the main line in the teaching process of each course, integrate relevant knowledge and skills into each link of the project, highlight the application of knowledge, guide students to think and innovate independently, and make the entire teaching form the project is the main line, the teacher is the leading, and the student is the main teaching model.

This project-oriented teaching method can well stimulate students' enthusiasm for learning and cultivate their independent learning, analysis and problem-solving abilities. In the whole process, the determination, completion of the project and the evaluation of the completion have become the central link of teaching. Teachers have changed from traditional "protagonists" to "supporting roles". Students no longer regard training and technical skills as boring learning tasks, but learn according to project requirements, changing from passively accepting knowledge to actively seeking knowledge.

$$r = tu^{y} \tag{3}$$

$$s = tv^{x} \tag{4}$$

The ultimate goal of dance art vocational education is that students can adapt to the market, be able to learn something and get a job on the stage where they need it. The traditional and closed "academic" teaching is often out of touch with the real society, and the content of students' learning does not match the needs of employment positions, which has caused a large number of students to find employment. The fundamental reason is that teaching is out of track with the market and the reality of employment. cause. In order to effectively solve this problem, in the process of exploring the innovative training model of talents, the dance major of Shanxi Vocational Art College highlights the occupational characteristics of the post, pays attention to the practical application of technical skills, and adopts the practical teaching mode of internship, through the combination of teaching and practice, so that Students have acquired the knowledge and abilities that they should have in their future jobs in specific artistic practice.

## III. EXPERIMENT

Image grayscale processing based on video action recognition technology is shown in the figure.

Fig.3. Image grayscale processing

The frame recognition effect of video dance moves is shown in the figure.

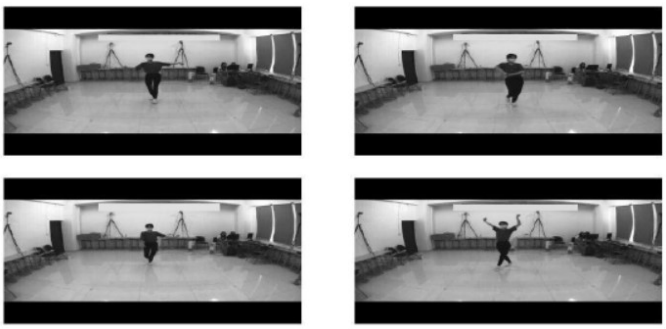

Fig. 4. Frame recognition effect

Based on video action recognition, the creative framework of dance teaching art is shown in the figure.

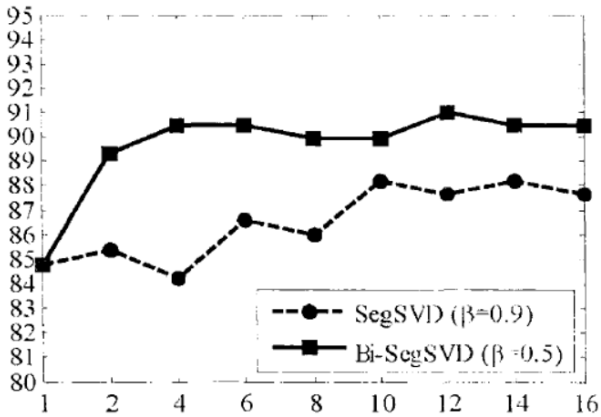


Fig. 5. Dance teaching art innovation framework

## IV. CONCLUSION

This article is mainly based on the video action recognition technology to study the artistic innovation framework of dance education. Simulation test results show that this method can effectively identify KTH and A-go-go dance video databases. First, it introduces the process, development and drawbacks of traditional dance education art; then analyzes the principles, algorithms and applications involved in motion recognition technology; finally, based on video motion recognition


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technology, studies the design plan of dance education art innovation framework.

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